

CO1-Assessment Tool-03

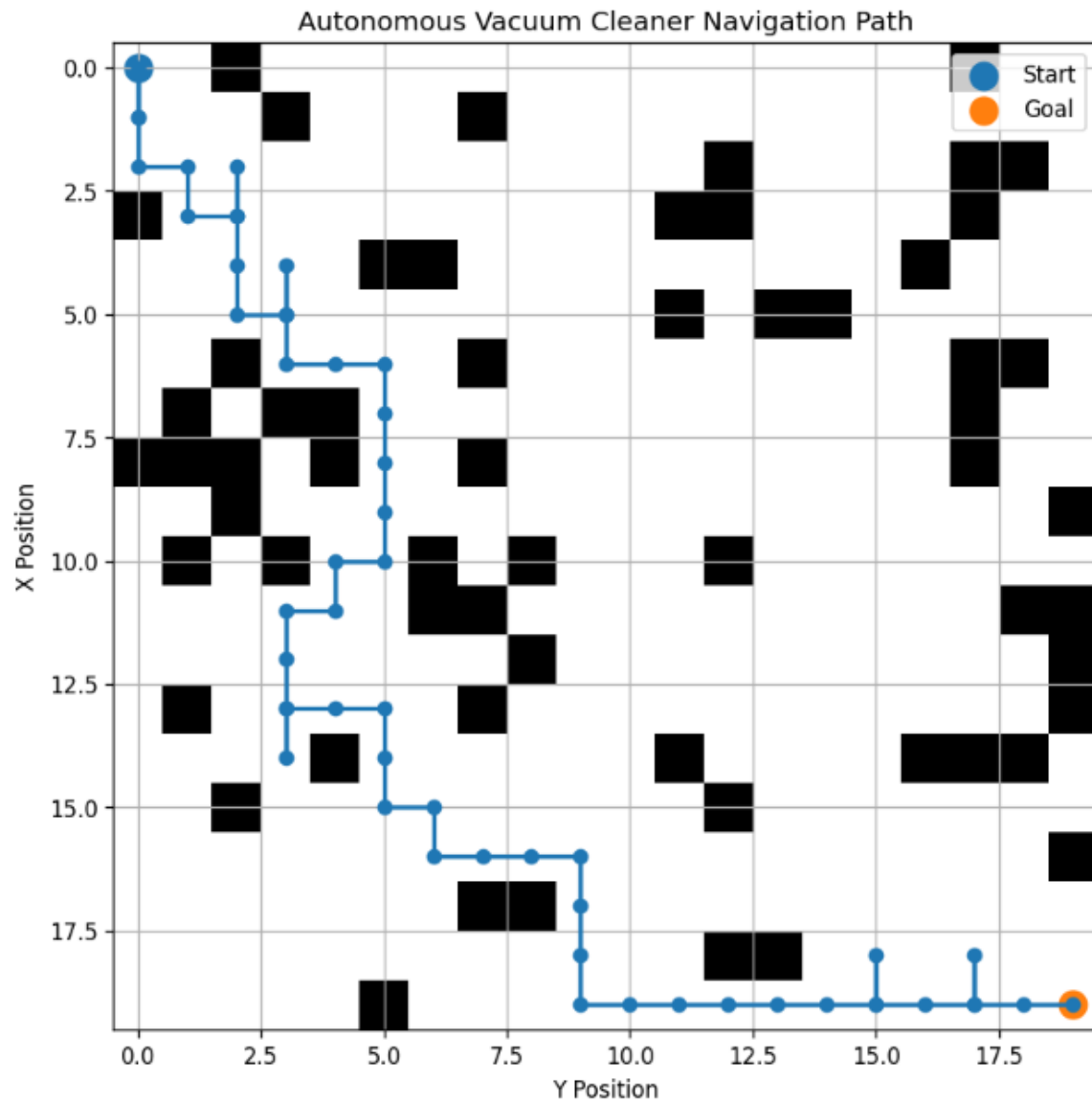
Reinforcement Learning

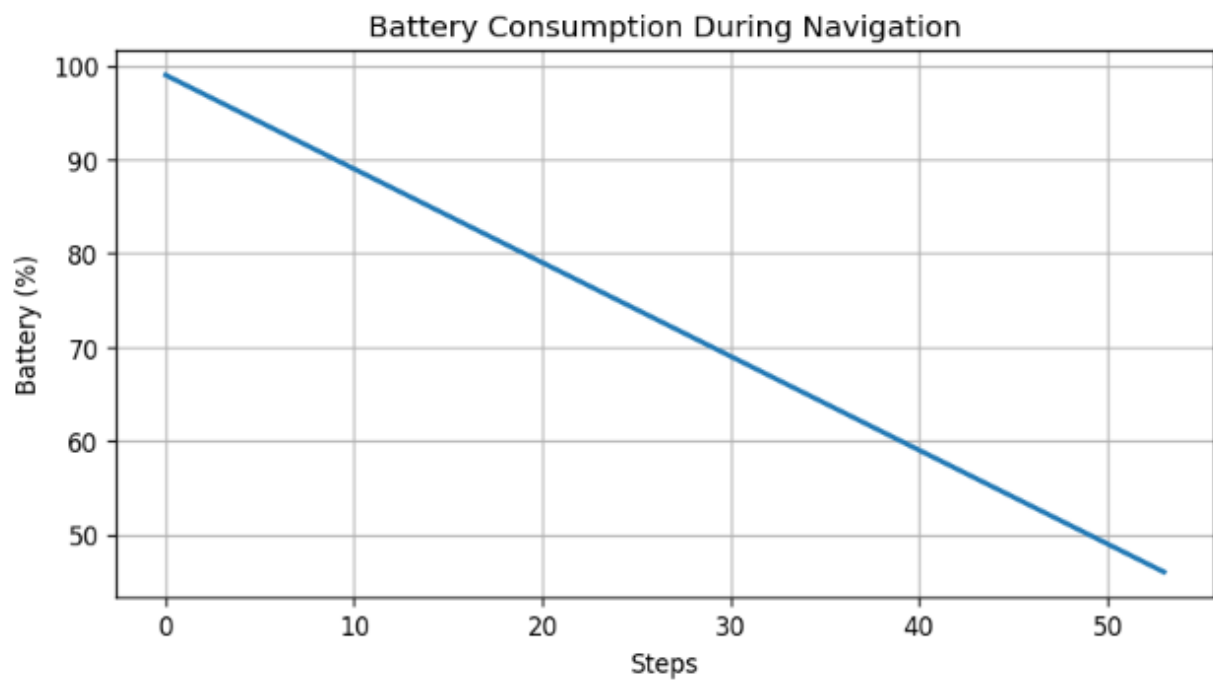
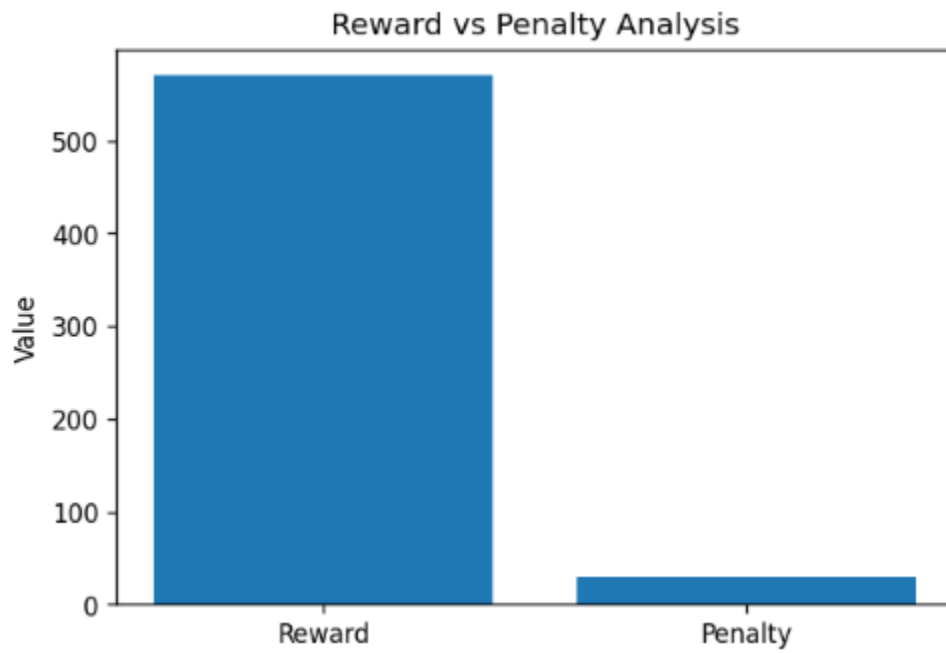
Name: -C.Jyothi

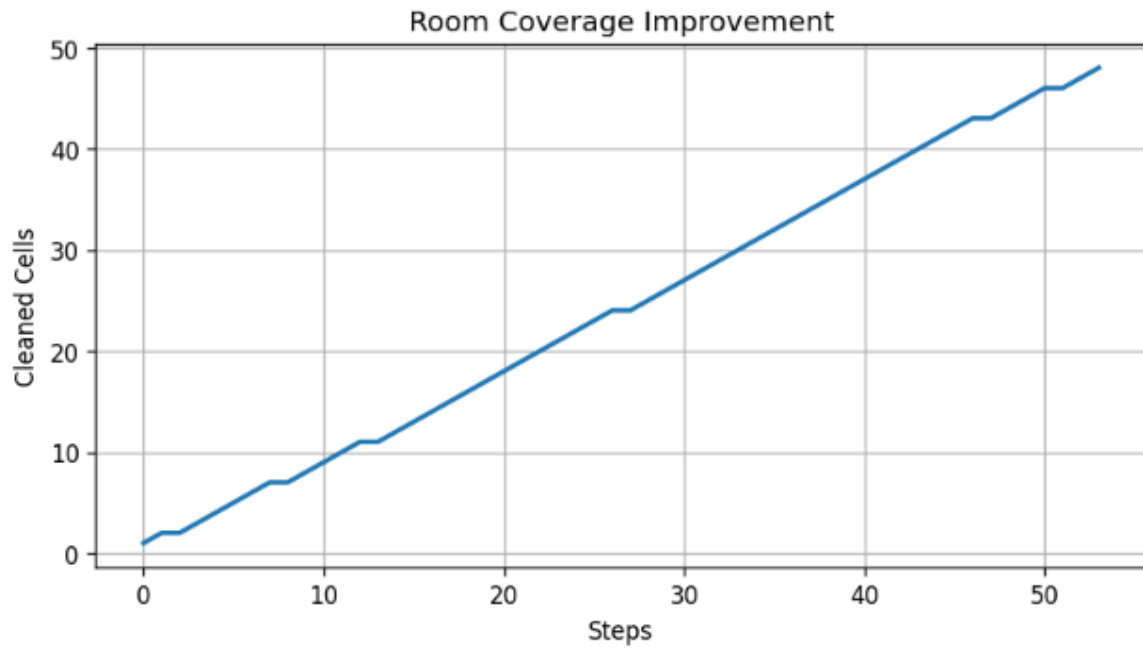
Regno: -192425312

Course Code: -MLA0305

Slot-B

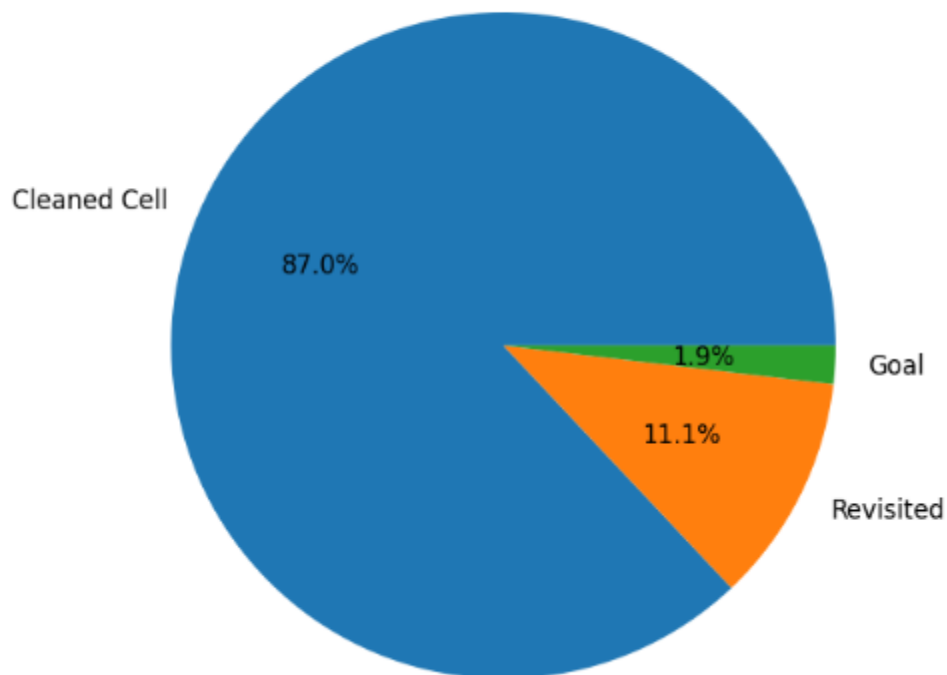


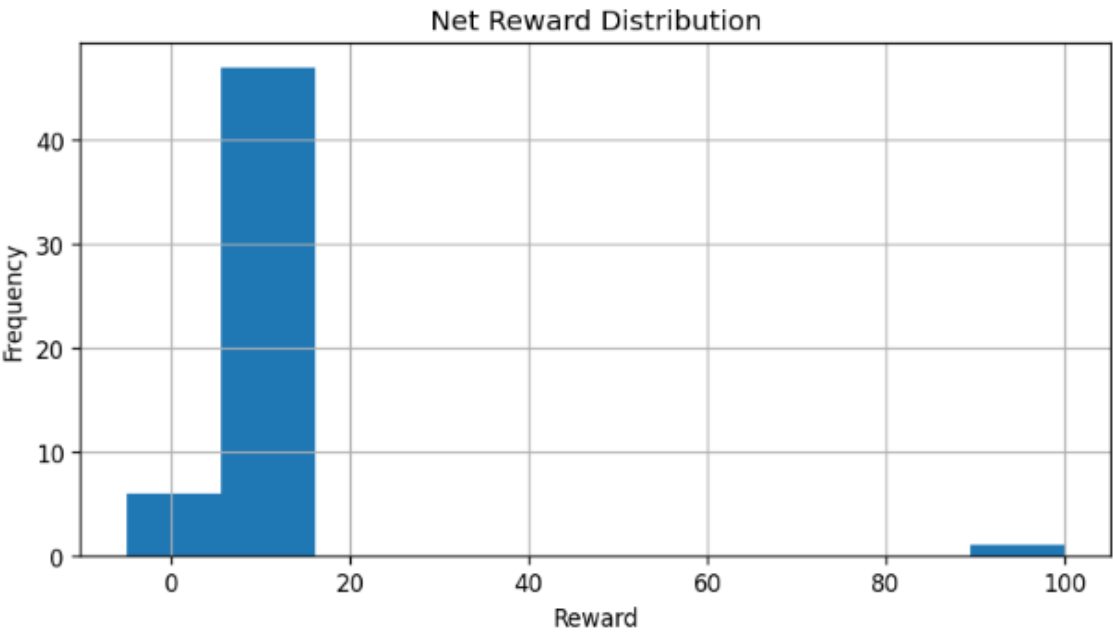




Navigation State Distribution

Navigation State Distribution





AUTONOMOUS VACUUM CLEANER PERFORMANCE SUMMARY		
	Performance Metric	Value
0	Environment Size	20 x 20
1	Total Simulation Steps	54
2	Coverage Cells	48
3	Coverage Percentage	12.0%
4	Total Reward	570
5	Total Penalty	-30
6	Net Reward	540
7	Obstacle Collisions	0
8	Boundary Violations	0
9	Revisited Cells	6
10	Battery Remaining	46%
11	Goal Status	Reached

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AUTONOMOUS VACUUM CLEANER SIMULATION DATASET

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	Step	X_Position	Y_Position	Action	State	Reward	Penalty	Net_Reward	Cumulative_Reward	Battery	Coverage
0	0	1	0	DOWN	Cleaned Cell	10	0	10	10	99	1
1	1	0	0	UP	Cleaned Cell	10	0	10	20	98	2
2	2	1	0	DOWN	Revisited	0	-5	-5	15	97	2
3	3	2	0	DOWN	Cleaned Cell	10	0	10	25	96	3
4	4	2	1	RIGHT	Cleaned Cell	10	0	10	35	95	4
5	5	3	1	DOWN	Cleaned Cell	10	0	10	45	94	5
6	6	3	2	RIGHT	Cleaned Cell	10	0	10	55	93	6
7	7	2	2	UP	Cleaned Cell	10	0	10	65	92	7
8	8	3	2	DOWN	Revisited	0	-5	-5	60	91	7
9	9	4	2	DOWN	Cleaned Cell	10	0	10	70	90	8
10	10	5	2	DOWN	Cleaned Cell	10	0	10	80	89	9
11	11	5	3	RIGHT	Cleaned Cell	10	0	10	90	88	10
12	12	4	3	UP	Cleaned Cell	10	0	10	100	87	11
13	13	5	3	DOWN	Revisited	0	-5	-5	95	86	11
14	14	6	3	DOWN	Cleaned Cell	10	0	10	105	85	12

